JOURNAL OF HEPATOLOGY

Normothermic perfusion of a human tumoral liver for 17 days with concomitant extracorporeal blood purification therapy: Case description

To the Editor:

Normothermic machine perfusion (NMP) is currently under extensive investigation for its ability to reduce graft injury and increase organ utilization. Moreover, NMP has the invaluable potential to be transformative as a platform for prolonged isolated organ studies and manipulation including physiology in deep analyses, therapeutics and gene product delivery.[1] In view of this, achieving stable long-term *ex vivo* homeostasis represents a fundamental prerequisite. In a landmark study by the Zürich group,[2] human livers discarded for transplant were normothermically perfused *ex situ* for 7 days, maintaining physiological parameters, by means of an *ad ho*c engineered prototype. Recently Lau *et al.*[3] prolonged for 13 days the *ex vivo* NMP of human discarded partial grafts and Hautz *et al.*[4] demonstrated that with the prolongation of perfusion the immune cell status shifts from a pro-inflammatory to an exhausted but also regenerative phenotype. Although these studies included a rudimentary dialysis system as a dialysis filter added in parallel for filtration of water-soluble toxins and regulation of perfusate volume, or to control the hematocrit by removing or infusing a specific solution in the circuit, to our knowledge, no NMP integrated with a pre- and post-dilution continuous hemodiafiltration technique has been described to date.

The NMP setting presented herein has been unprecedentedly integrated with an organ extracorporeal blood purification system to maintain a physiological condition of organ perfusion over time, not only in terms of oxygenation and CO_2 removal, but also in waste product removal, while preserving the normal range of pH, electrolytes, and glucose. To satisfy these purification requests we developed a novel translational platform integrated with blood purification to obtain stable long-term homeostasis in NMP-perfused human livers. The model takes advantage of commercially available elements ready to be integrated in a multifunctional platform. In addition, to overcome the issue of limited numbers of discarded livers as an experimental organ source, we developed a resource strategy that provides for the use of diseased human livers explanted during transplantation for long-term *ex situ* perfusion studies and research.

We herein report an unprecedented case of prolonged NMP for 424 hours (17 days) of a liver obtained at the time of transplantation from a 56-year-old patient with unresectable colorectal liver metastases. After hospital ethical committee approval and adequate patient informed consent, the liver of the patient affected by colorectal metastases was explanted as the first step of liver transplantation with a technique resembling that used in autotransplantation cases. The hepatectomy technique was slightly adapted to avoid damage to liver inflow vessels for subsequent cannulation and to reduce warm ischemia time. Recipient safety and oncologic radicality were always overriding in considering technical choices. The explanted liver was immediately connected to a commercially available NMP (Liver Assist ®XVIVO) without any cold flush (WIT 8 min).

To achieve long-term perfusion and guarantee a close-to-physiological liver environment, the system was integrated with a long-term oxygenator (Eurosets, New Born, Modena, Italy) and with an extracorporeal blood purification therapy (EBPT) system (Fig. 1A). In this platform the temperature was set at 37 °C and maintained for the whole perfusion time that started after electrolyte, hematocrit and pH recovery in the blood through the EBPT. Oxygen delivery was provided by an air/O_2 mixture (800 ml air, 35-40% FiO_2).

Perfusion was maintained for 17 days (424 hours) with good functional and homeostatic parameters throughout the whole period except for the last 24 hours. Perfusion was terminated after a progressive increase in cholestatic parameters and decline of liver function starting from day 15. Both portal vein (PV) and hepatic artery (HA) perfusion flows and resistances were stable throughout the entire long-term perfusion with minimal para-physiological variations (Fig. 1B,C). Liver functional activity was tested by lactate clearance, synthesis of coagulation factors and factor V, injury markers, maintenance of albumin levels, and blood urea nitrogen production (Fig. 1D,E). Bile production started at 6 h after perfusion and continued until the end of the experiment with an average value of about 30±7 ml/day. Bile analysis showed glucose and bilirubin clearance into bile; the electrolyte content overlapped with that measured in the perfusate. Tissue injury assessed by histological analysis (H&E) at day 13, day 15, and at the end of perfusion demonstrated preserved architectural integrity of the liver with no evidence of substantial cell death (Fig. 1F). As far as cytokine analysis is concerned, a peak of IL-10 was measured at 2 hours from the beginning of perfusion while the other inflammation markers remained within standard values over the whole perfusion. CEA (a molecule with a half-life in physiologic conditions of 7 days and mainly metabolized by the liver) value was monitored throughout the whole perfusion; it remained measurable over time and at a concentration like that measured in the patient before transplantation A broad-spectrum antimicrobial agent was used prophylactically and strict aseptic handling was applied. Perfusate cultures taken at day 13 and at the end of perfusion were negative.

The case presented herein is a proof of concept on an innovative perfusion model with the potential for maintaining organs in prolonged normothermic machine perfusion for more

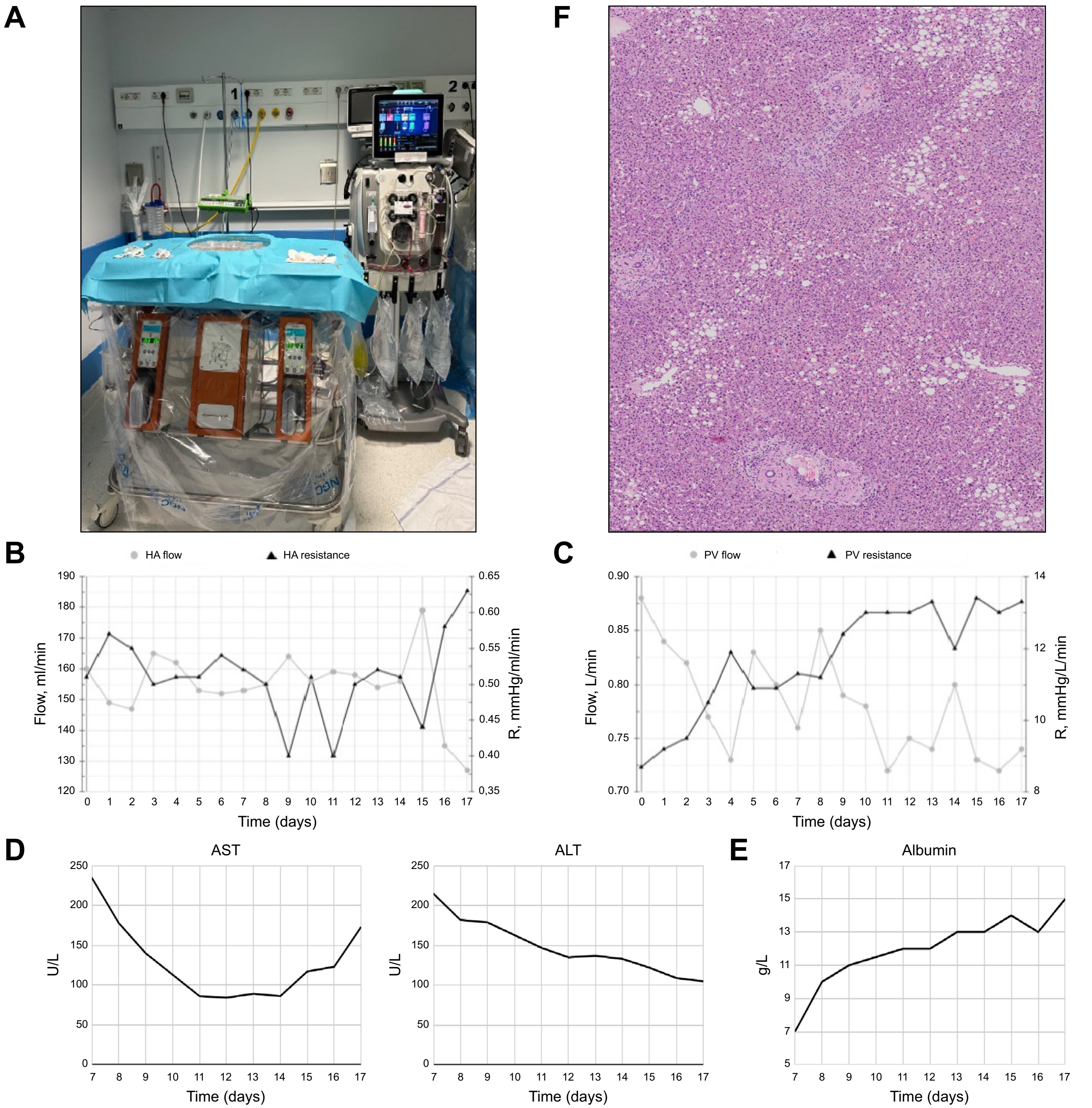

Fig. 1. Liver extracoporeal perfusion system and function parameters. (A) Liver extracorporeal perfusion system combined with extracorporeal blood purification therapy. (B) HA flow and resistance. (C) PV flow and resistance. (D) Perfusate analysis AST and ALT. (E) Perfusate analysis albumin. (F) H&E staining of liver at the end of perfusion. ALT, alanine aminotransferase; AST, aspartate aminotransferase; HA, hepatic artery; PV, portal vein.

than 2 weeks. Such a time span is long enough to warrant manipulations compatible with many biological processes, such as regeneration, inflammatory down modulation, genotypic and phenotypic cell changes. The system that allowed for successful liver NMP of more than 2 weeks was achieved by adopting a long-term oxygenator, an EBPT efficiently integrated with a commercially available machine perfusion device, nutritional support and antimicrobial therapies. The integrated blood purification system was not merely constituted by a filter unit but by a complex monitor for continuous renal replacement therapy in the hemodiafiltration configuration, with personalized dialysate and replacement fluids. An artificial kidney was integrated into the system to maintain a physiological environment and represents a key element for platform success in reaching homeostasis during very long-term perfusion. Crucially, all the integrated elements are "on market" devices already approved for clinical use and commercially available: this could allow for easier and faster clinical translation.

The additional innovative element that characterizes this experiment and differentiates it from other published works on long-term perfusion is the use of a human organ obtained from the recipient's hepatectomy during transplantation. Adopting the total hepatectomy technique developed for auto-transplantation, we showed that using a diseased liver after hepatectomy for NMP experimental purposes is feasible without any risk to the patient and led to perfectly preserved long-term liver function. Furthermore, this was a diseased organ affected by foci of colorectal metastases: the presence of organ tumors creates a unique and relevant research model and offers tremendous potential for future studies in liver pathology and cancer in a controlled *ex vivo* environment. Furthermore, this technology provides a valuable platform for liver-related basic research in an *ex vivo* setting, facilitating investigations on novel therapeutic interventions and drug testing.

Herein, we report on protracted (>15 days) human liver NMP made possible by an integrated platform including a blood purification system. The quality of the homeostasis obtained throughout most of the experiment is promising and, if confirmed, paves the way for future *ex vivo* organ manipulation interventions including gene therapy interventions, such as those directed to achieve "immunologic invisibility" for organ transplantation and immunomodulation. The model could also serve as a tool for pro-regenerative studies with the aim of achieving parenchymal regeneration in an *ex vivo* setting. Furthermore, this model offers an unprecedented setting for investigating the pathophysiology of "diseased livers" such as those affected by colorectal metastases. The utilization of commercially available components makes this innovative human model more readily translatable into clinical practice.

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Received 2 April 2024; Received in revised form 16 April 2024; Accepted 17 April 2024; Available online 2 May 2024
https://doi.org/10.1016/j.jhep.2024.04.024

[†]Shared first authorship.

Financial support

Progetti di Ricerca Bando SID 2022 (delibera n.149/2022) "Setup of a translational model of *ex vivo* normothermic prolonged perfusion of human explanted organs as experimental platform for drug testing and gene therapy: a pilot study to assess feasibility and cell vitality".

Conflict of interest

The authors declared that they do not have anything to disclose regarding conflict of interest with respect to this manuscript.

Please refer to the accompanying ICMJE disclosure forms for further details.

Authors' contributions

UC conceptualization, resources, supervision, writing-review and editing; FN methodology, investigation, supervision, writing-review and editing; AB methodology, validation, investigation, data curation, writing-review and editing; SI validation, writing-review and editing; EG methodology, investigation, supervision, writing-review and editing.

Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jhep.2024.04.024.

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